

## **Brewers Baseball Ops Roundtable**

Student manager events and baseball conferences helped people who set up event

Be collaborative

## **GM of Brewers**

Executive of the Year Winner

Be curious, ask questions, and be a good teammate

## **Wake Forest Data Analyst**

College War - adjust based on quality of conference

Plans to make projection system

### Hitting cWAR:

Park factor adjusted

wOBA based formula

Hitting, defense, and baserunning

Underestimates extremes of baserunning and defense

### Pitching cWAR:

AVG of FIP and ERA

Park factor adjusted

Leverage index multiplier applied to relievers

Wake Forest ranks transfer portal based on cWAR

## **Another Wake Forest Analyst**

Batted ball outcome model

In the future, compare the 2 models to see insights into defense, calculate OAA, and use pitch shape and type, number of outs, and count to predict outcome

Around 80 ABs to stabilize wOBA and xwOBA

Hyperparameter tuning was hard

Doesn't really take into account defensive talent rn

Think it would be pretty applicable at MLB level but infield would probably take a hit

Predict batted ball outcomes using solely ball flight data from Trackman - xwOBA:

College baseball data using database from Trackman and TruMedia data and omitted errors

XGBoost Model with 5 features: exit speed, bearing, and 3 others

Stratified by batter side

xBA and xSLG were very close to BA and SLG values

Predict batted ball outcomes using ball flight data from Trackman and player positioning:

About 5% increase in accuracy.

xBA and xSLG were off less than 10 points

### **Dartmouth Student**

Worked with Ishan Kinikar and Levon Siriah

When to pull a starter? Won 2024 SMT Data Challenge

Used minor league data

Used 2023 expected runs data for MLB to make difficulty metric

Completely anonymized data so didn't totally factor strength of other teams hitters

Could add handedness variable

Runs wasn't factored in

xPLG - expected pitches left in the game:

Most important factor: pitches in game

xPLI - Expect pitches left in the inning:

Most important factor was difficulty, then pitches in inning, then pitches in game

Snell Tool app - input confidence and fragility then it determines when to pull pitcher

Minor leaguers pitch 4 extra pitches based on stat he made

## **VP of Domestic and International Scouting, Brewers**

5 years he was a baseball intern, california collegiate league he started in

AVG number of intern years is probably 2-3 years

Started with Brewers as a minor league video intern in 2017

Saw Joe Adell and other major leaguers

Did a similar role in Helena, Montana

Worked on video projects for international scouting in offseason

R and D intern in summer 2019 for international scouting

Transitioned to international scouting department

Learned international scouting from home without traveling (it was March 2020)

Need to be very good at acquiring players for Brewers to be competitive

Late nights since play at night, nocturnal industry

Bus rides across the Rockies

Learned from players, different life experiences from high schooler vs college player

He learned a lot from coaches: baseball wisdom, how to improve players, life experiences, etc.

From scouts learned their routine, what they're looking for in batting practice

Learned how other teams are evaluating players

Acquired players in 2018 that fueled NLCS run

Seek out own interests - learn through applying code to that, better than learning in classroom

Learn unique circumstances that influence baseball players around the world

Can apply your interests in different job roles

Likes asking people in an interview how they strategically think about non baseball stuff, crazy stuff, high school fantasy drafts and investing in stocks. Interviews about 60+ people per year.

Non quantitative stuff is important, area scouts are following their gut feel

Scouting information is a driving predictor in how good players will be. Works in harmony with analytics

He's always learning

Seeing someone in person helps see how strong and big a player is, hard to tell on video

## **Director of Baseball Analytics and Video at Archbishop Muller High School**

Master of Business Analytics at Purdue

Ken Griffey Jr and Barry Larkin went to high school

Pitcher Success Formula, Attack Zone success, MoeStuff+, Pitcher Success ML Model, Dashboard Navigation Demo, Moeller Pitcher RShiny Dashboard

Use pitch type and location

15000 pitches using AWRE

Pitcher success formula: pitch result: ball is -1, 1.2 for swing and miss, at bat result, count leverage, and location modifier (points for each)

MoeStuff+: points based on at bat results, slider could be given 140 rating and fastball could be 120

XG Boost model with cross validation for pitcher success model:

Binary features based on two strike counts, is late inning, etc.

Top 3 pitcher combinations

Dashboard: [idbach16.shinyapps.io/MoellerPitcher2025V2/](http://idbach16.shinyapps.io/MoellerPitcher2025V2/)

Will have some Trackman data and can add Rapsodo data. Can make it more predictive.

About 10 people came together and took votes and made a lot of changes for the point values

A lot of trial and error

On Dashboard:

Probability of each pitch being successful for different pitchers, MoeStuff+, AVG MoeStuffPlus by pitcher, top 2 pitcher pairs

## **2024 SMT Data Challenge**

Route Optimization, WAR Monger posts about it

Catch probability model and route optimization model from anonymized minor league data

Catch probability model:

Random Forest model

Route Efficiency Model:

Used fielder speed and catch probability model and distance to ball bounce/catch to make optimized path

OAA is flawed cuz it doesn't give players credit for increasing the catch probability from start of the play

Route score gives players credit for taking a good route even if he didn't catch the ball

Wants to add a give up point when catch probability is low and look at AVG fielder speed on the season rather than on the route

Limited storage on Shiny app, Random forest model had less storage

## **Recent Graduate of University of Northern Colorado with Masters in Applied Stats**

Investigating the Problem of Bases Empty OBP

OBP is higher with runners on, higher BABIP with runners on

Defense can position better with bases empty

How can hitters increase walks with bases empty?

Got over 800 individual seasons with at least 400 PA from 2021

Predicted uBB% with Oswing%, Contact%, Zone%, EV, and more

Predicted K% with Zone%, OSwing%, and more

Used foul ball strike rate in both

If you take in 2-0 and 3-1 counts in bases empty, OBP should increase but SLG less. BB% should increase

Suggests singles hitter take in 2-0 and 3-1 count, while power hitters swing away

### **Senior VP of Player Development and Performance, Brewers**

Was pitching coach in minors, Coached at college level; Played college baseball

Head coach at alma matter

You're never really prepared for the next job

Put force into the ground when pitching

Tries to fit what player does well along with optimizing for best results

Take 6 weeks to build you up if you haven't played competitively in 6 weeks

Draft minicamp tells players they signed the resources they have

Make complex info simple, get it to players by understanding how players learn (have psychology group) and having relationships with them

Educate coaches to help them handle follow up questions from players

Low A pitching coach he thinks feels connected to front office

Coaches need to make information actionable and digestible

Try to build skills that scale to big leagues

Care more about things that lead to success than ERA and batting average (swing at perfect pitch and hit it hard)

Try to get players to fall in love with work that it takes to get results

Care about individual pitches' Stuff+ and want to throw best stuff pitches in high leverage

Can't make total arsenal better without making each pitch better

Can have 40 out of 80 arsenal but still be successful using art of pitching

## **Brewers Executive Advisor to the GM**

Every decision in baseball comes down to player evaluation is what college coach told him

Started with Rockies in small front office as an intern, led to a lot of projects given to him

Worked in the Labor Relations Department for commissioner's office - saw how all 30 teams operated, small vs big market team and analytical vs scouting driven team. Met CEO of a team, who became president of baseball ops for the Orioles. He then worked for Orioles who rebuilt

Worked for Angels who were win now. The guy who he met who became president of baseball ops for the Orioles then interviewed him for the Phillies GM job and he got that. Helped them become more modern (data driven more)

Wants to work with good people who challenge you and a forward thinking organization - found match with Brewers. Wanted to learn

A lot of colleagues that are very talented here at Brewers

Collaboration better here than any other organization

Be helpful in acquisition process, communicate well, and be a good teammate - want best for team and help teammates out

Wants to see how players do in game situations, their athleticism, etc

Be open-minded to what may come next

Don't believe your way is the best so you keep having the hunger to learn

Player health is next competitive advantage, especially with pitchers

winning culture flows from Leaders - lead by example

Really hard to force feed culture, based on personnel, develops as it develops

Day never really ends, when you go to bed, work still isn't done - as GMs say not enough hours in the day so surround yourself with people you trust. No typical day

Everything is rooted in evidence/objectivity

## **Recent grad of University of Oregon and West Coast League worker**

### **MLB Draft Signing Bonus Model:**

XG Boost - can handle nonlinear interactions and feature importance

Separate player by hitter vs position and perceived talent

Wants to add more direct prediction of player skill, quantification of health history, and more data and grade alignment

Would like to have CI to see prep guys vs college guys

## **Iowa Baseball Analysts or Former Analysts**

### **Predictive Player Evaluation:**

Stickiness - How well does a metric predict itself in the future

Stuff is more predictive of run value and sticky than location

Stuff predicts well quickly while Quality (stuff and location) is your best once you get enough pitches (100s). Location can vary a lot in small samples

### **Flash Shapes:**

Shows movement velo and other stuff inputs to show players what they've already done and get them to throw certain pitches more

True Talent stat - regress stat based on number of pitches seen

Keep models interpretable so players and coaches know what goes into model

## **Major League Game Preparation Specialist**

Works under player info department

Mostly offensive focused

Did MLB Pipeline program - cool dinner with Tyrone who introduced staff, 6am classroom session on scouting, then hot fields of Phoenix, watch BP and games, learning session, and learn how to write reports. Wrote scouting reports on players (wrote 5 reports a night probably). Arguing with people who they like vs dislike. Saw Mason Miller and Jackson Merrill

Hybrid between scouting and major league strategy

Plays first base and shags fly balls

You don't have all the answers

Played in high school cuz Sal Frelick got injured and played against some major league players

Quit last 2 years of college baseball to work with Brewers



What can you do to help the team win today?

High level if about to play team for the first time, how has specific player changed since last time we played team if playing team again

Pitch arsenal and ability to locate for scouting reports for hitters

Stay for every home game if you work for a team

Did a lot of data cleaning for the draft

Learned technical skills on the job - codecademy and ChatGPT

Maybe break down report by monthly, tries to use large sample

Simple visuals, visuals hold most weight

2 sentence summary

Goes over gameplan for starter and bullpen themes to hitting coaches

Hitting meeting every day where he'll go over specific bullpen guys, about something specific like someone throwing more strikes, to start a conversation with each other

Coaches and players are normal people

Don't insert yourself out of emotion, sleep and be positive tomorrow

Camaraderie, there is relationship, even if something goes bad on field like fight or being lazy

Things may get misconstrued to media

Have players get out and practice on weird fields

Communicate weirdness of ballparks

Be authentic and prepared (have some relevant info) in interviews. If you're fake you may get the job, but the real person will come out during job and you probably won't keep job. You don't need a ton of questions, turn it into a conversation. Reach out and thank people.

Work Life balance - more reliant on coffee from letting self go, affected work, do little things to make yourself better

Among Intern group, people worried about standing out, but don't do something that will undermine someone. Collaborate and be willing to work with other departments. Have a normal conversation with coworkers. Be willing to ask questions.

### **Closing Remarks with 2 Brewers Employees who Set this Up**

Don't train in the zoo, train in the jungle (challenging situation)

Growth mindset

Put we before me

Person that gets hired is who we think can fit into our culture, not who has the best resume

Be okay with failure inherent to job search not just baseball

Improve yourself from questionnaires and ask for feedback

Intangibles are important - hard worker and good teammate and projects that show how you can add wins is more important than resume

Points to tangible work someone did when hiring

Answering front office question in project is very valuable

MLB approved this event

### **Full Time R and D Analyst and 2 Interns (Interactive Session)**

Majored in Data Science and interned with Phillies for 2 years and worked with Iowa Baseball as Data Analyst

Intern with Tigers, University of Michigan, worked with Rays, and works with Brewers now

quantitative finance after college and worked in baseball in college,

Name, where you went to school, baseball experience, and where you want to go

**On a questionnaire - there are not right answers but there are wrong answers - want people to think about baseball the right way - can back up answers with a good thought process. The questionnaire is a very big portion of the final decision. Show creativity and competency. Do you know what you're talking about and are you up to date on the state of the game? In the modeling part, deal with feature selection and missing values and be creative.**

Write cover letter

Submit some code and modelling questions

After questionnaire 1 to 4 virtual interviews - baseball and technical questions in those

Know the team you're interviewing with - know the best players, shows passion

Simpler models for why and care about inputs and player development

More complex model if you just care about prediction

Full time R and D staff travel to an affiliate every year

Every homestand minor league can ask analyst questions

In spring training talk to coaches

Don't act like you know everything, talk to pitching coach about things that would be important for predicting pitcher performance

Communicate in their terms, if into batting average then put wOBA on batting AVG scale. It's hard to get out of their lingo. Be confident when communicating or coaches won't listen.

If a couple things go your way, that helps coaches to buy in.

Old school coaches think about things the same way just don't articulate it the same

Make plots self explanatory, coaches not all used to graphs

Convey information to coaches is the job

Pick and choose your fights, don't fight a lot if it doesn't affect win probability a lot

Pretty open to info at Brewers

Think about why they won't use your suggestions (maybe they have subjective information)

Typically, coaches won't turn a blind eye to convincing information.

Day to day is built around an overarching project then things come up from coaches and others. 75% or more of day from Python or R. 25% of day on Slack and in meetings.

Director of R and D reviews code, does less code, is in meetings

A lot of really hard work done by entry level employees

At 7pm can go out and watch team play

Interns - SQL and R for day to day, ad hoc stuff, random questions on project, and then watch games. Work feels meaningful.

Data cleaning is a big challenge

Hardest part is when projects get the trickiest

More poor data quality likely in international and amateur - names not spelled correctly

Data engineers try to make data clean

Very close with data engineers

Career arc for R and D - entry level, lead/senior analyst then maybe become head of baseball ops or assistant GM or director of R and D

A lot of employees started as Brewers interns

Apply to internships

Intern said he's glad he's an intern and not full time

Building predictive ML models is a big part of the job

Most teams probably use R, but should still get job with Python

Public models are similar to what teams use

Scale data from different sources is a challenge, data engineering helps

Qualitative aspects are hard to quantify

Well rounded is good, caution limiting yourself to 1 area

Be really knowledgeable/expert on what you've done

Teams may bucket people into pitching, hitting, fielding, etc.

Get target variable right for projection, make necessary adjustments

Don't shy away from EDA and sample size limitation

Use all data for projection (maybe even 7 years), then adjust based on what model says, can say 7 years ago is useless

Limit arbitrary decisions

Final advice: watch baseball, think of what player will do and why players do certain things, and can lead to creativity